



Recent insights into pesticide resistance mechanisms in *Plutella xylostella* and possible management strategies

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Abstract

Insects are incredibly successful and diverse organisms, but they also pose a significant threat to agricultural crops, causing potential losses of up to US\$470 billion. Among these pests, *Plutella xylostella* (Linnaeus), a devastating insect that attacks cruciferous vegetables, alone results in monetary losses of around US\$4–5 billion worldwide. While insecticides have effectively protected plants under field conditions, their use comes with various environmental and mammalian hazards. Additionally, insects are developing resistance to commonly used insecticides, rendering management strategies less effective. Arthropods employ a range of behavioral and biochemical mechanisms to cope with harmful chemicals, which contribute to the development of resistance. Understanding these mechanisms is crucial for addressing the issue of resistance. It is imperative to integrate strategies that can delay the development of resistance and enhance the efficiency of insecticides. Therefore, we present an overview of insecticide resistance in insects, focusing on *P. xylostella*, to provide insights into the current resistance status of this pest and propose tactics that can improve the effectiveness of insecticides.

Keywords Insecticide resistance · Pest management · Resistance mechanisms · Integrated pest management · Insect pest resistance

Introduction

Insects are one of the most diverse, successful living entities, encompassing about three-fourths of the total organisms on earth (Grimaldi et al. 2005). Besides their beneficial aspect (Akhtar and Isman 2018), many insects pose significant

challenges as major pests of agricultural commodities (Mueller et al. 2005) and as invasive species (Siddiqui et al. 2021). Estimating actual losses caused by insects is challenging due to the variability influenced by environmental conditions, cultivated plant species, socioeconomic factors, and technological capabilities (Dhaliwal et al. 2015). Global estimates emphasize that insects are responsible for pruning

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approximately 18–20% of crop production, resulting in losses exceeding US\$470 billion (Oerke and Dehne 2004).

Like other arthropod pests, several Lepidopterans, including *Plutella xylostella* (Linnaeus) (Lepidoptera: Plutellidae), commonly known as the Diamondback moth (DBM), are expanding their geographical range (Oliveira et al. 2014). DBM is a significant insect pest that has garnered considerable attention for study. The exact geographic origin of DBM is still a subject of debate. This insect's widespread distribution can be attributed to several factors, such as short generation time, broad host range, ability to survive adverse environmental conditions, and seasonal migratory behavior (Chapman et al. 2002; Perry et al. 2018; Li et al. 2021). Chemical control has been widely used, offering quick knockdown effects and minimizing yield losses when alternatives are lacking (Nansen and Ridsdill-Smith 2013). However, the use of insecticides has varied over time and space since World War II (Sharma et al. 2019). The dilemma arises from the injudicious and consistent use of a limited number of similar chemicals, leading to significant selection pressure and the evolution of resistant populations (Enayati et al. 2020; Siddiqui et al. 2023). Compounding the issue is the development of cross-resistance against insecticides with different modes of action, further exacerbating the situation (Kostromyska et al. 2018; Hafeez et al. 2022a; Wang et al. 2021). Moreover, the DBM can rapidly develop resistance against various insecticide groups (Khakame et al. 2013).

To effectively manage insect damage, it is crucial to analyze the underlying mechanisms contributing to the emergence, spread, and persistence of resistance in insect pests. Studying insecticide resistance provides valuable insights into resistance's ecological and evolutionary aspects, which can guide the development of insecticide resistance management (IRM) strategies through an integrated approach. By understanding the mechanisms involved, we can aim to prevent or mitigate the development of insect resistance. This review aims to provide an updated overview of the resistance status of *P. xylostella* and offer potential recommendations for its management. By examining the current state of resistance and exploring management tactics, we can enhance our understanding of this pest and propose effective strategies to tackle the challenges.

Origin and distribution of *Plutella xylostella*

The origin of the Diamondback moth (DBM) has been debated among researchers. One argument suggests that DBM originated in Europe due to the high number of Lepidopteran pest species found in that region compared to others. However, some scientists believe it originated in Africa (Talekar et al. 1993). Another hypothesis suggests that DBM originated in America, as it was initially reported in 1854, indicating a possible origin in this region (Philips et al.

2014). However, a recent You et al. (2020) study using conserved mitochondrial gene analysis, specifically cytochrome Oxidase-I (COI), suggested that South America may be the initial habitat for *P. xylostella*. This finding was supported by the analysis of isolated mitochondrial genes from China and pairwise comparisons, indicating the migratory nature of the DBM in Asia (Wei et al. 2013). The DBM has achieved global notoriety as a major insect pest of cruciferous crops, persisting throughout the year (Furlong et al. 2013). However, there is evidence that DBM cannot overwinter in western Canada's and northern Japan's temperate regions, leading to sporadic invasions in these areas (Honda 1992; Dosdall et al. 2001; Miluch et al. 2014).

Host plants and damaging potential

Brassica is an important genus within the Brassicaceae family, comprising 37 species cultivated for various purposes (Sanlier and Guler 2018). However, globally, *Brassica oleracea* and *Brassica rapa* are well known for their health benefits and are extensively consumed (Manchali et al. 2012; Sanlier and Guler 2018). The DBM, being a generalist feeder, attacks almost all members of the *Brassica* crops. However, its preferred hosts include broccoli, cauliflower, cabbage, brussels sprouts, oilseed rape, kohlrabi, reddish, and turnip (Furlong et al. 2013; Shehzad et al. 2021). In addition to cultivated *Brassica* crops, there are several cruciferous weeds, such as wild mustards like charlock mustard (*Sinapis arvensis* L.), which are annual plants primarily found in Europe (George et al. 2009). Another weed, *Erysimum cheiranthoides* (L.), is commonly found along the roadside and serves as a significant harbor for *P. xylostella* throughout the year (Sarfraz et al. 2011). Shepherd's purse *Capsella bursa-pastoris* (L.) Medicus, native to North America but prevalent in Asia and Europe, also provide refuge for the DBM (Muhamad et al. 1994; Sarfraz et al. 2012).

P. xylostella is responsible for approximately 90% of defoliation in *Brassica* crops, attacking them from seedling to maturity stages (Ayalew 2006; Machekano et al. 2020). It has been identified as one of the most economically damaging pests of *Brassica* crops (Li et al. 2021). In China alone, the economic losses caused by *P. xylostella* amount to approximately \$0.77 billion (Li et al. 2016). A single outbreak in California resulted in a loss of 6 million US dollars (Shelton et al. 2000). The widespread distribution of this pest makes it challenging to estimate the total damage accurately. However, Talekar and Shelton (1993) estimated a monetary loss of US \$1 billion in 1992. Australia has experienced sporadic infestations of this pest since the 1800s, resulting in significant crop losses (Furlong et al. 2008). In Europe, *P. xylostella* continues to be a devastating

pest for crops. In western Canada, a single infestation in 2001 led to treating approximately 1.8 million hectares of fields with insecticides (Zalucki and Furlong 2011). Globally, US\$4–US\$5 billion were lost due to direct losses and management costs of *P. xylostella* (Zalucki et al. 2012). The larval stage of *P. xylostella* is responsible for crop damage through defoliation, resulting in skeletonized leaves (Moreira et al. 2016). Additionally, the presence of live and dead larvae in broccoli and cauliflower heads can lead to market rejection of the produce (Farias et al. 2020; Shen et al. 2020). Given the alarming threat posed by this pest, prompt management is crucial, and chemical control remains the preferred strategy at the field level.

Resistance status of *Plutella xylostella*

Due to the short life cycle and significant genetic heterogeneity of DBM strains, controlling DBM using treatment techniques is challenging (Zago et al. 2014; Bacci et al. 2019). In modern agriculture, the use of insecticides is essential to meet the increasing demand for food and ensure early protection of crops from pest species (Lengai et al. 2020).

Several factors influence farmers' inclination toward insecticide usage on crops:

- 1- A favorable growing season with adequate rainfall and other favorable conditions can lead to high crop yields but also poses a threat of increased insect pest attacks, which compels farmers to use chemicals to maintain crop health.
- 2- Farmers' investment in seeds and fertilizers is another factor that lowers the threshold for tolerance imposed by insect pests, making using insecticides more necessary (Nansen and Ridsdill-Smith 2013).

Pesticide usage has been increasing over time. In 2012, producers globally spent nearly \$56 billion on pesticides (Atwood and Paisley-Jones 2017). According to a pesticide marketing report 2019, insecticide sales reached \$85 billion, projected to grow at a compound annual growth rate (CAGR) of 11.5%, surpassing \$130.7 billion by 2023 (Pesticide Report 2023). The economic impact of *P. xylostella* infestations is significant, resulting in direct losses and management costs of approximately US\$4–5 billion. These losses underscore the substantial investment, totaling billions of dollars annually, dedicated solely to controlling *P. xylostella* populations (Zalucki et al. 2012). The indiscriminate use of pesticides has a detrimental effect on the environment, mammalian health, and the development of resistance in *P. xylostella* (Shakeel et al. 2017; Junaid et al. 2023).

The earliest documented case of resistance in *P. xylostella* was reported in Java, Indonesia, in 1951 against dichlorodiphenyltrichloroethane (DDT). It was observed that a higher

quantity of insecticide was required compared to the previous year (Ankersmit 1953). Since then, *P. xylostella* has consistently ranked as one of the top 20 most resistant species of insects, having developed resistance to more than 97 insecticides (Table 1).

In order to develop a more effective strategy to cope with insecticide resistance and manage the insect pest, it is crucial to understand the mechanisms contributing to evasion against certain chemicals (Scott 1990; Liu 2015). Investigators have reported the following mechanisms of resistance development.

Basic mechanisms in resistance development

1. Cuticular resistance (penetration resistance)
2. Behavioral resistance
3. Metabolic resistance
4. Target site alteration

The mechanism by which resistance evolves depends on the species of insect as well as the geographical origin of the species (Kliot and Ghanim 2012; Ullah et al. 2016).

Metabolic resistance

The biochemical transformation of a toxin into an innocuous form is termed metabolic resistance (Khan et al. 2020a). Biochemical transformation frequently occurs in two intervals; phase-I involves detoxifying xenobiotics, which is carried out by redox and hydrolytic reactions. The phase-II mechanism works in conjunction with phase-I and poisonous chemicals being applied. The preliminary phase involves several hydrolytic and redox reactions, leading to the detoxification of xenobiotics. The second phase of metabolism consists of the biosynthesis of conjugative enzymes that bind to toxins or primary products of phase-I, making them more water-soluble and facilitating their elimination from the body (Fig. 1) (Zhou et al. 2019). Cytochrome P450 monooxygenases (P450s) are a group of phase-I metabolic enzymes critical in transforming endogenous and exogenous substances (Veith and Moorthy 2018). On the other hand, glutathione S-transferase (GST) is a phase-II metabolizing enzyme that catalyzes the reaction of reduced glutathione with compounds bearing electrophilic sites. This process results in the formation of more water-soluble metabolites that can be easily excreted from the body (Fig. 1) (Mak and Denisov 2018; Hafeez et al. 2022a, b, c; Khan et al. 2021; Khan et al. 2020a). Fluctuations in the concentration of cytochrome P450s

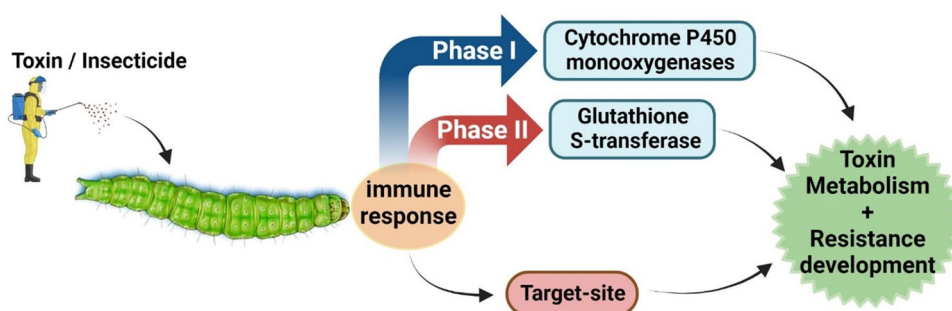
Table 1 Resistance reported in *P. xylostella* from the globe

Insecticide	Type of resistance	Country	Level of resistance	Citation
Leufenuron and Methoxyfenozide	Field evolved	Brazil	High level	Arruda et al. 2020
Indoxacarb, Abamectin and lufenuron	Field evolved	Brazil	moderate to high resistance	Santos et al. 2011
Chlorpyrifos	Field evolved	China	low to moderate	Wang et al. 2022
Indoxacarb	Field evolved	China	low to moderate resistance	Wang et al. 2022
Chlorantraniliprole	Field evolved	China	moderate to high resistance	Wang et al. 2022
Esfenvalerate and Permethrin	Field evolved	Australia	Moderate level	Baker and Kovaliski 1999
Deltamethrin, Bifenthrin and Lambda-cyhalothrin	Field evolved	Pakistan	Moderate to high resistance	Sayyed et al. 2005a, b
Abamectin	Field evolved	China	High level of resistance	Pu et al. 2010
Emamectin benzoate	Field evolved	China	High level of resistance	Pu et al. 2010
<i>Bacillus thuringiensis</i> (bt)	Laboratory based	China	Moderate level	Liao et al. 2019
<i>Bacillus thuringiensis</i> (bt)	Field evolved	USA	Moderate level	Ballester et al. 1999
<i>Bacillus thuringiensis</i> subsp. <i>Kurstaki</i>	Field evolved	USA	Moderate level	Tabashnik et al. 1990
<i>Bacillus thuringiensis</i> subsp. <i>aizawai</i>	Field evolved	USA	Low level	Shelton et al. 1993
<i>Bacillus thuringiensis</i> subsp. <i>Kurstaki</i>	Field evolved	USA	Moderate	Shelton et al. 1993
<i>Bacillus thuringiensis</i> subsp. <i>Kurstaki</i>	Field evolved	USA	Cross resistance	Tabashnik et al. 1993
<i>B. thuringiensis</i> subsp. <i>aizawai</i>	Field evolved	USA	Cross resistance	Tabashnik et al. 1993
<i>Bacillus thuringiensis</i> <i>kurstaki</i> Toxin Cry1Ac	Field crop	Malaysia	Coss resistance, Cry1Aa, Cry1Ab, Cry1Fa; Cry1Ca	Sayyed et al. 2004
<i>Bacillus thuringiensis</i> (<i>Bt</i>) insecticidal Cry proteins	Field crop (transgenic canola)	Malaysia	Cry1Ac	Sayyed et al. 2004
Insecticidal protein and commercial formulation	Cry1Ac, <i>Bacillus thuringiensis</i> var. <i>kurstaki</i> (Btk), Cry1Ba	China	Low to moderate	Wang et al. 2007
<i>Bacillus thuringiensis</i> var <i>kurstaki</i> (Biobit®)	Laboratory condition	India	Cry1Aa, Cry1Ab, Cry1Ac, Cry2A and Cry2B δ -endotoxins	Mohan and Gujar 2000
<i>B. thuringiensis</i>	CryIA,	Philippines	Moderate	Ferre et al. 1991
<i>B. thuringiensis</i>	CryIB, CryIC	Philippines	Low level	Ferre et al. 1991
<i>B. thuringiensis</i>	<i>B. thuringiensis</i> subsp. <i>Kurstaki</i>	US	Moderate	Perez and Shelton 1997
<i>B. thuringiensis</i>	<i>B. thuringiensis</i> subsp. <i>Aizawai</i>	US	Lower	Perez and Shelton 1997
Deltamethrin	Field evolved	Pakistan	Higher level	Sayyed et al. 2005a, b
Indoxacarb	Field evolved	Malaysia	Higher level	Sayyed and Wright (2006)
Indoxacarb	Field evolved	China	Moderate level	Zhang et al. 2017
Chlorantraniliprole	Field evolved	Brazil	High level	Ribeiro et al. 2017
Chlorfenapyr	Laboratory evolved	Brazil	low level	Oliveira et al. 2011
Spinosad	Laboratory evolved	Brazil	Low level	Oliveira et al. 2011
Spinetoram	Laboratory evolved	Brazil	Moderate	Oliveira et al. 2011
Diamide	Field evolved	Philippines	Moderate level	Steinbach et al. 2015
Diamide	Field evolved	Philippines and Thailand	Moderate level	Trocza et al. 2012
Diamide	Field evolved	China	High level	Wang et al. 2013
Diamide (spintoram)	Field evolved	India	High level	Tamilselvan et al. 2021
Chlorfenapyr	Laboratory evolved	Brazil	Moderate	Lima and Siqueira 2017

have been associated with developing resistance to pyrethroid insecticides, while glutathione S-transferases are involved in detoxifying organochlorine compounds. Esterases and oxidases primarily act on carbamate compounds (Fig. 1) (Enayati et al. 2005; Balabanidou et al.

2016; Pavlidi et al. 2018; Hafeez et al. 2023). Upregulation of GSTs has been reported in several resistant *P. xylostella* populations at both genomic and proteomic levels. The GSTu1 gene promotes chemical degradation by masking the binding site, while the PxGST2L

Fig. 1 Schematic diagram of the metabolic resistance (chemical metabolized before reaching to the target site)



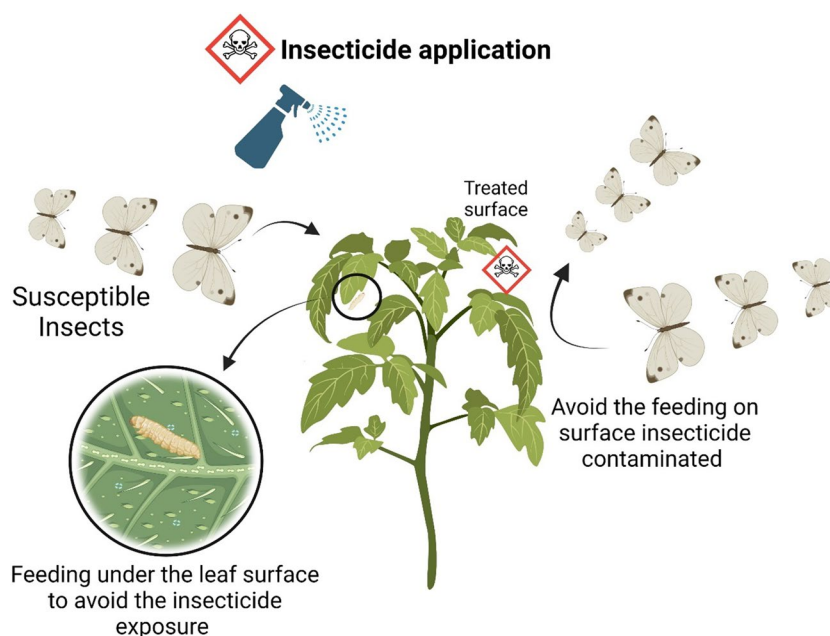
protein reduces toxicity by decreasing GST activity in the case of chlorantraniliprole resistance, respectively (Yin et al. 2021; Zhu et al. 2021). Stress caused by Acephate, Indoxacarb, beta-cypermethrin, and Spinosad induces the regulation of GST-related genes, resulting in observed upregulation and downregulation of PxGSTe3, PxGSTs2, and PxGSTd3, respectively, making DBM immune to these insecticides (Chen and Zhang 2015). Detoxification enzymes and the cytochrome P450 complex have also been associated with resistance in *P. xylostella*. The overexpression of several genes, such as CYP321E1, CYP340s, and CYP6BG1, leads to increased production of cytochromes, enhancing insecticide detoxification and rendering DBM immune (Hu et al. 2014; Yu et al. 2015; Li et al. 2018). Cross-resistance has also been linked to Increased CYTS production (Fig. 1) (Yu et al. 2015). Furthermore, researchers have discovered that certain microbes present in the bodies of insects play a critical role in detoxifying insecticides (Siddiqui et al. 2022a).

Behavioral response of *Plutella xylostella*

Resistance studies play a pivotal role before formulating management and resistance mitigation strategies to reduce the outbreaks of *P. xylostella* and related pests. There are several mechanisms by which insects adapt to a particular chemical, and behavioral response is the preliminary and vital one (Haynes 1988; Siddiqui et al. 2023). Insects detect danger and start avoiding it, often ceasing feeding when encountering a certain hazardous chemical. The pest may also avoid treated areas by relocating to adjacent untreated fields or deeper crop canopies (Fig. 2) (IRAC 2022; Roode and Lefèvre 2012). Multiple studies emphasize behavioral modifications in *P. xylostella* in response to insecticide.

According to reports, *P. xylostella* has developed behavioral resistance, as evidenced by the ability of adults to choose their egg-laying sites and the mobility of larvae to avoid lethal doses of foliar-applied pesticides on crops (Zago et al. 2014; Sarfraz et al. 2005). Silva Filho et al. (2020) detected differences in the locomotive activity of *P. xylostella* adults

Fig. 2 Schematic diagram of behavioral resistance, including avoidance and feeding at the non-treated parts of the host plant



in the populations from the field and laboratory. The field-collected populations showed increased walking time and reduced resting interval compared to the control, allowing the insects to escape from treated surfaces. Insecticides do not solely drive evolution in insect behavior. A wealth of evidence suggests that host plant evasion also plays a role in developing resistance. For instance, insects may exhibit resistant features by depositing more eggs or selectively choosing plants with less hairy and sensitive cuticles for oviposition. This behavioral adaptation enables them to enhance their survival and reproductive success in the presence of various challenges (Silva and Furlong 2012).

Penetration resistance

In this resistance mechanism, insects modify their cuticles by depositing specific hydrocarbons or altering the composition of their integument. This modification reduces the penetration of pesticides into their bodies, slowing down the inward flow of pesticide molecules and delaying their interaction with target proteins within neuronal cells (Fig. 3) (Balabanidou et al. 2016; Fang et al. 2015). While this resistance mechanism has been extensively studied in insect vectors such as mosquitoes and bed bugs, fewer studies have been conducted on cuticular resistance in *P. xylostella*.

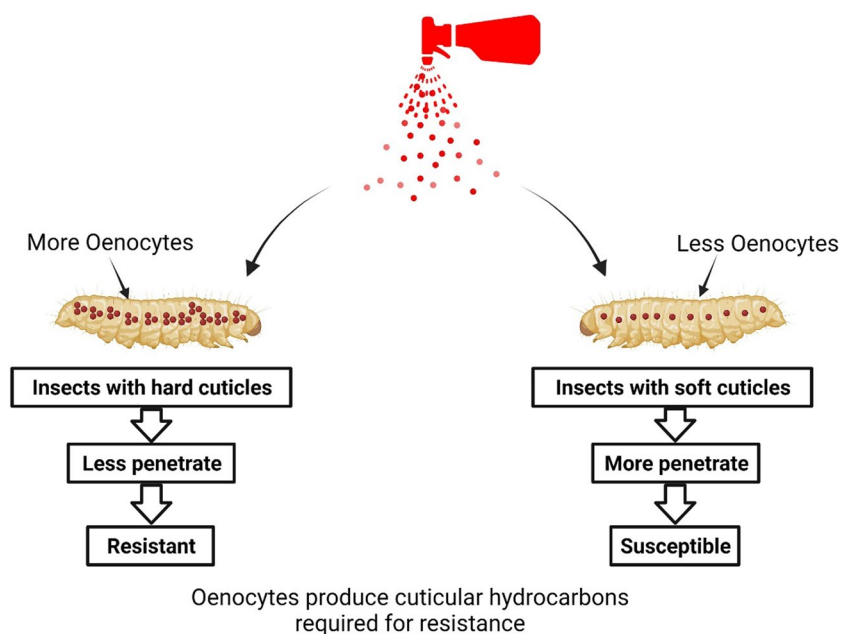
A preliminary study by Noppun et al. (1989) observed that immatures with reduced cuticular penetration rates exhibited higher excretory rates, indicating a correlation between reduced penetration and resistance. Zhu et al. (2017) found significant differences in dermal microRNAs

between a chlorantraniliprole-resistant population of *P. xylostella* and a susceptible population, providing further evidence of cuticular modification during resistance development. Similar findings were reported by Shabbir et al. (2021a, b), showing the upregulation of factors involved in cuticle formation, such as dehydrogenase and chitinase, in resistant strains of DBM. Other related insects have also shown signs of cuticular resistance. For instance, Ahmad et al. (2006) demonstrated that *H. armigera* effectively deterred deltamethrin impregnation due to thicker cuticles compared to susceptible individuals, where pesticide penetration was significantly higher throughout the body. Ngumbi et al. (2020) reported fluctuations in the hydrocarbon profile of the cuticle when insects were exposed to repetitive insecticide pressure, suggesting a potential role of cuticular resistance in *Amyelois transitella* (Lepidoptera: Pyralidae) (Fig. 3).

Target-site resistance

Each group of insecticides targets specific sites to exert their pesticidal effects (Silva and Furlong 2012). Through repeated exposure to a particular group of insecticides, insects with high adaptive behavior can develop mutations in the target sites, thereby increasing their survival rate (Boaventura et al. 2020; Hafeez et al. 2022a). Target site mutations leading to resistance are specific to certain groups of insecticides. For example, acetylcholinesterase (AChE) serves as a target for carbamates (CBs) and organophosphates (OPs) by disrupting AChE functioning, which in turn

Fig. 3 A schematic diagram of the cuticular resistance mechanism



disrupts nervous transmission, and eventually causes insect death (Siddiqui et al. 2022b, c).

Numerous incidences have been reported where alterations in acetylcholinesterase confer resistance in DBM strains (Rinkevich et al. 2010; Baxter et al. 2010). Studies on AChE resistance in DBM and related insects have observed mutation in amino acid sequences that influence the expression of *ace1* and *ace2* genes, which code for AChE, leading to insect insensitivity (Lee et al. 2007; Kim et al. 2012). Mutations in the amino acid sequence have been reported at several points, such as A298S and G324A in the case of acephate resistance, and three mutations in the prothiofos-resistant strain (Sonoda and Igaki 2010; Sindhu et al. 2018; Lee et al. 2007). Guo et al. (2017) found a positive association between the G227A mutation and resistance to organophosphate or carbamate insecticides. Sonoda et al. (2014) identified multiple haplotypes of AChE encoding genes in field-collected populations compared to susceptible ones. Several studies have emphasized that the mutation in the AChE gene is reversible and can decline if insecticide pressure is reduced (Lin et al. 2017).

Ryanodine receptor mutation

Ryanodine receptors (RyR) are specific channels that facilitate the release of calcium ions during muscle contraction (Van Petegem 2012). While three different RyR isoforms have been characterized in mammals, only one isoform has been identified in invertebrates (Sattelle et al. 2008). The most recent group of insecticides, known as diamides, target these channels for their effectiveness (Truong and Pessah 2019). The first case of resistance to diamide insecticides in *P. xylostella* was reported in 2011 (Srinivasan et al. 2011; Troczka et al. 2017). It was later discovered that resistance

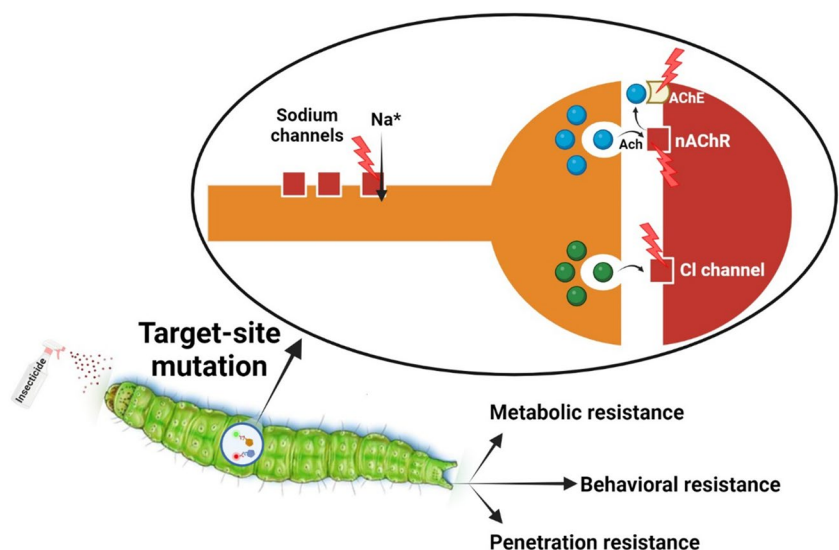
was due to an amino acid substitution at position 4946 in the RyR region, where glycine was replaced with glutamic acid (Troczka et al. 2015; Guo et al. 2014). Subsequent studies, such as Kang et al. (2017), confirmed that a point mutation (G4946E) in the RyR gene is responsible for high resistance in *P. xylostella* against chlorantraniliprole. The G4946 mutation has also been observed in other insect pest species, including *Chilo suppressalis*, *Tuta absoluta*, and various *Spodoptera* species (Yao et al. 2017; Sang et al. 2016; Roditakis et al. 2017). Further research revealed another mutation known as the I4790K mutation, where the isoleucine amino acid is substituted with lysine at position 4790 (Jouraku et al. 2020).

Voltage-gated sodium channel

Voltage-gated sodium channels are proteins that play a crucial role in the transmission of electrical signals in nervous cells (Corry and Thomas 2012; Siddiqui 2022b, c) (Fig. 4). Due to their essential function in nerve signal transmission, they serve as a target for numerous insecticides (Silver et al. 2014). The mode of action of insecticides like DDT and, more recently, Pyrethroids is based on the prolonged activation of sodium channels, which delays the normal voltage-dependent inactivation (Tian et al. 2018).

Insects have developed mechanisms to become insensitive to these insecticides, known as knockdown resistance or *kdr* (Rinkevich et al. 2013). *Kdr* resistance was first identified in the housefly, and later super-*kdr* was characterized in housefly and several other insects (Busvine et al. 1951; Du et al. 2013). The A309V gene is involved in regulating chloride channels in nerve transmission. Mutation in this gene has been observed to decrease the sensitivity of *P. xylostella*

Fig. 4 Overview of target-site mutation mechanisms in *P. xylostella*



to abamectin by inhibiting glutamate gates responsible for chloride ion movement (Wang et al. 2017).

ATP-binding cassette transporter (ABC)

ABC proteins constitute a family of proteins present in the midgut of insects and serve as binding sites for specific microbial agents, particularly Bt (Schneider and Hunke 1998; Locher 2009). Decreased expression of these receptor genes leads to reduced effectiveness of Bt. The reduced expression of these receptor genes hampers the interactions between the toxin and receptors, thereby facilitating the development of Bt resistance in various insects.

Comparison of different resistance mechanisms

It is worthwhile to discuss the prevalence comparison of different resistance mechanisms, although each occupies its rank and is interplayed. In the context of DBM, resistance to pesticides, penetration, and metabolic resistance are essential mechanisms. However, their prevalence, importance, and relevance can vary depending on the specific situation and the type of pesticide being used. Penetration resistance refers to the ability of DBM to reduce the entry of pesticides into its body through physical or physiological mechanisms. This can involve adaptations such as thicker cuticles or altered entry points that hinder the pesticide from reaching its target site. On the other hand, metabolic resistance involves the efficient detoxification or breakdown of pesticides by DBM, reducing their toxic effects. The relative importance of penetration and metabolic resistance can differ based on factors such as the pesticide's mode of action and the specific resistance mechanisms present in the DBM population. In some cases, penetration resistance may be more prevalent and relevant, impeding the pesticide's effective targeting. In other cases, metabolic resistance may dominate, enabling DBM to metabolize the pesticide and reduce its effectiveness.

Metabolic resistance is commonly observed in insects and is considered the most prevalent resistance mechanism. Studies, such as Adesanya et al. (2019), have shown that selection for physiological resistance to acaricides in *Tetranychus urticae* (spider mites) resulted in altered behavioral responses, suggesting that behavioral modalities can be modified following physiological alterations. Similar findings can be applied to *P. xylostella*, where biochemical resistance involves the modulation of enzyme activities (Gong et al. 2013). Target site insensitivity, involving modifications in the insect's cuticle or digestive tract linings to impede insecticide absorption or penetration, is another

common mechanism observed in some strains of resistant insects.

To summarize, while both penetration and metabolic resistance are relevant in DBM resistance to pesticides, the prevalence and importance of each can vary. Metabolic resistance is the most common mechanism observed in insects, followed by target site insensitivity. However, the specific prevalence and relevance of these mechanisms can be influenced by the type of pesticide and the resistance profile of the DBM population under investigation.

Tactic to reduce insecticide resistance in *P. xylostella*

Farmers cannot completely eliminate the need for insecticide application; however, it is possible to reduce the likelihood of pesticide resistance development (Gill 2014). Several tools have been employed in this regard.

Mechanisms to manage insecticide resistance

Crop monitoring/insect pest (Economic Threshold Level (ETL))

The first and most important strategy that should be implemented to delay resistance development is the proper monitoring of crops to determine the Economic Threshold Level (ETL) of an insect pest. The ETL represents the pest density at which control measures should be applied to prevent the increasing pest population from reaching the economic injury level. It is essential to use insecticides only when the insect population reaches a point where their presence poses a significant risk of economic losses that outweigh the cost of pesticide application or when there is a threat to public health (Pedigo et al. 2021; IRAC 2022). Model-based studies have demonstrated that establishing and adhering to the economic threshold can have profound implications for effectively managing pesticide resistance strategies (Peck and Ellner 1997).

Insecticidal rotation

The basic concept of insecticide rotation is simple: reduce the insecticide pressure of a particular toxin by using an alternative toxin for a limited period of time. South and Hastings (2018) demonstrated that an insecticidal mixture is the best alternative for preventing insecticide resistance in mosquitoes. According to IRAC, the key principle for delaying resistance is to avoid repeated exposure to insecticides with the same mode of action against the target insect

(IVCC 2022). This approach reduces the selection pressure on a specific group of insecticides, thus delaying resistance (Ahmad et al. 2002). In the case of *Helicoverpa armigera* (Hübner) resistance in cotton, effective management was achieved by using different groups of insecticides in succession (Russell et al. 1998). Similar findings were reported by Ahmad (2004), highlighting that the mosaic model of insecticide application can hinder the development of insecticide resistance. Therefore, ongoing efforts are focused on the discovery of newly formulated insecticides with preferably new modes of action and improved environmental profiles to combat the development of insecticide resistance. Rotation programs have also been successfully implemented to delay selection pressure against *P. xylostella*. In Georgia, resistance in *P. xylostella* was delayed by alternating the use of Bt, chitin inhibitors, sodium channel blockers, and diamides (Riley and Sparks 2009).

Mixture of insecticides

Fine-scale insecticide combinations have been considered essential methods for resistance control in many insect pests (South and Hastings 2018) (Fig. 5). Theoretically, it is believed that applying a mixture is more effective for resistance management because the evolution of resistance to individual insecticides in the mixture is primarily rare. Therefore, there is a lower likelihood of resistance development to both insecticides simultaneously (Curtis 1985). Model-based results from Sudo et al. (2017) support the robustness of a mixture strategy for managing resistance, provided that initial resistance is low and refuge supplies are available. The blended insecticidal strategy has been successfully tested on a substantial number of insects from different orders, such as aphids (Taillebois and Thany 2016), *H. armigera* (Martin et al. 2003), *Spodoptera litura* (F.) (Ahmad et al. 2009) and house flies (Khan et al. 2013). This strategy has also been implemented successfully against DBM. Attique et al. (2006) found a synergistic effect of a selected insecticidal mixture against a resistant population of *P. xylostella*. Another study by Ahmed et al. (2022)

discovered a synergistic effect of methoxyfenozide when applied along with piperonyl butoxide against a selected insecticidal population. Recently, Shabbir et al. (2021a, b) demonstrated the synergistic effect of Bt and chlorantraniliprole against selected strains of *P. xylostella*.

Before applying a mixture of insecticides, it is crucial to consider the specific mode of action associated with each insecticide. It is highly recommended to use pesticides with different modes of action to maximize effectiveness and minimize the risk of resistance development (Buss et al. 2007). Failing to address the mode of action adequately could lead to unfavorable outcomes (<https://pesticidestewardship.org/>). The Insecticide Resistance Management Committee has provided a mode of action scheme to assist in making informed decisions. This scheme serves as a valuable resource for government and state agencies, advisors, consultants, universities, growers, and extension staff, offering guidance on the selection of insecticides and acaricides as part of a cyclical or alternating strategy for pest resistance control (Sparks and Nauen 2015).

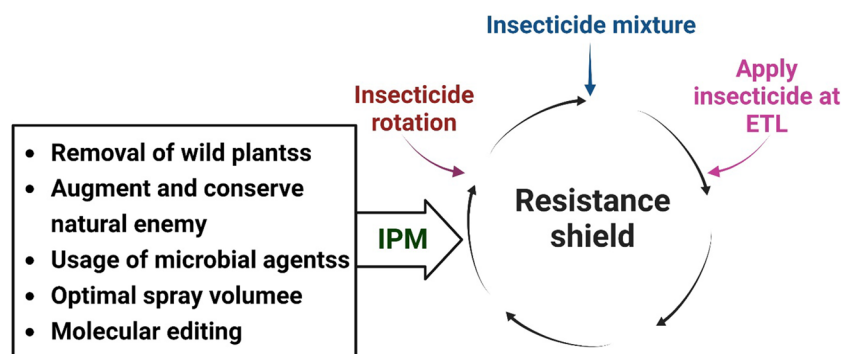
Genome-based approaches

Insect genome editing has shown promise as a successful technique for resistance management. It involves genetically modifying target insects and reintroducing them into their natural environment (McFarlane et al. 2018). The development of insensitivity to certain chemicals can be attributed to changes at receptor sites, either at the genetic or molecular level. Genome editing utilizes the concept that an organism can regain susceptibility if the altered factor is modified. Recently, the CRISPR/Cas9 editing technique has been used to reverse an allelic-based mutation associated with *kdr*-type resistance in fruit flies (Kaduskar et al. 2022).

Use of mixed cropping

The mixed cropping patterns of the usage of refuges are recognized as a crucial strategy for delaying pest resistance. This strategy has demonstrated successful implementation

Fig. 5 Factors that can contribute toward the breakage of resistance shield in an insect



in delaying resistance to Bt products in various insect species (Tabashnik et al. 2008; Zafar et al. 2020). The primary objective of this strategy is to impede the inheritance of resistance genes in *P. xylostella* by providing refuges of non-Bt host plants (Tabashnik et al. 2008).

Several studies support the effectiveness of this approach. Huang et al. (2014) observed a reduced oviposition rate of *P. xylostella* in mixed cropping systems, indicating alterations in host-finding behavior, which subsequently reduces pesticide load and resistance development. Additionally, Liu et al. (2014) put forth a compelling argument suggesting that resistance development can be delayed by implementing refuge broccoli plants alongside natural enemies such as *Coleomegilla maculata*. These findings collectively support the notion that refuges, combined with mixed cropping and the presence of natural enemies, can contribute to resistance management in *P. xylostella* populations.

In summary, the implementation of refuges or mixed cropping patterns is an essential strategy for delaying resistance in *P. xylostella*. Studies have demonstrated that this approach can disrupt the heritability of resistance genes, reduce pest oviposition rates, decrease pesticide load, and potentially enhance the efficacy of natural enemies. Further research in this area would be valuable to refine and optimize the use of refuges for effective resistance management in *P. xylostella* populations.

IPM-based program

Monitoring is the first and crucial step in an integrated resistance management toolbox. This step involves monitoring the pest population and gathering knowledge about their susceptibility to insecticides, tracking resistance trends, and understanding the mechanisms of resistance (IRAC 2022; Stenberg 2017). Integration of artificial intelligence tools, such as detection surveys, pest risk maps, and hotspot analysis, has become essential (Siddiqui et al. 2023). Nowadays, biomarker-based strategies for insecticide resistance monitoring and RNAi-based insecticides are also important considerations for resistance management (Zhu et al. 2017). The utilization of insect pheromones is another tool for early monitoring, and trapping or mating disruption programs can help reduce the reliance on insecticides (Tacain et al. 2016).

Various pest control methods can be used in agriculture, including synthetic or biological insecticides, beneficial insects, cultural practices like crop rotation, transgenic or pest-resistant crop varieties, chemical attractants or deterrents, and microbial agents. Combining these strategies can effectively manage pest populations while minimizing risk (Siddiqui et al. 2022a). It is advisable to incorporate more tactics than relying on a single one to manage resistance. This approach reduces the selection pressure on resistant alleles, leading to a quicker reduction in resistance (Alphey et al. 2009; Bamisile et al. 2021a, b; Khan et al. 2020b).

IPM models have been developed to delay resistance in *P. xylostella*. The study by Harvey-Samuel et al. (2015) found that releasing low-level releases of MS (male-selecting transgene) males effectively inhibited population growth and delayed the spread of Bt resistance when combined with Bt broccoli that expressed Cry1Ac. Neither MS nor Bt alone could achieve this outcome. An efficient IPM model begins before crop sowing, where the initial step is to plant varieties that have resistance characteristics against specific insects. In the case of DBM, varieties with higher levels of glucobrassicin, glucoiberin, and glucoiberin are less likely to be attacked by *P. xylostella*, reducing the need for pesticide applications (Robin et al. 2017). Integration of biological control methods and resistant varieties, combined with timed cultural practices, significantly reduces pesticide usage and thereby delays the development of insecticide resistance (Liu et al. 2014) (Fig. 5).

Conclusion and future concerns

In conclusion, the extensive research conducted on the development of resistance in *P. xylostella* provided valuable insights into the rapid evolution of resistance traits. It has been observed that resistance primarily arises through mechanisms such as boosted metabolic detoxification or alterations in target sites. Over the past four decades, the control of *P. xylostella* has heavily relied on three main chemical classes. Carbamates and organophosphates share a similar mode of action, while pyrethroids target voltage-gated sodium channels, which have shown a decline in their effectiveness against the pest. Although the introduction of neonicotinoid insecticides initially reduced concerns, recent developments have demonstrated that DBM has also developed tolerance mechanisms to these chemicals.

Given these challenges, there is a pressing need to introduce new chemicals with different modes of action to combat resistance in *P. xylostella*. Additionally, biocontrol practices should be implemented effectively, utilizing methods such as parasitoids or entomopathogens to enhance pest management strategies. By diversifying control measures and combining chemical and biological approaches, we can address the evolving resistance in *P. xylostella* and ensure sustainable pest control in the future.

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Declarations

Competing interests The authors declare no competing interests.

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